Assumption-Gated LLM Agents for Continuous-Treatment Difference-in-Differences

An Application to E-commerce Delivery Times

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ABSTRACT

How much does a slower delivery cost an e-commerce platform in future sales? Delivery time is a continuous treatment, and practitioners answer such questions with a generalized propensity score (GPS) combined with difference-in-differences (DID). We show that the variant common in industry—a normal GPS with the Hirano–Imbens two-step estimator—is biased by a factor of 2.5–3 and produces confidence intervals that cover the truth in only 12–14% of simulated replications when delivery times are right-skewed and heavy-tailed, as they are in practice. A cross-fitted doubly robust dose-response estimator on a heteroskedastic gradient-boosting GPS, with an undersmoothed bandwidth and a subsampling variance inflation, restores nominal coverage (96.5%). We wrap this estimator in an LLM agent whose numbers come only from a deterministic engine and whose every step passes an executable gate: covariate timing, weighted balance, overlap, a placebo parallel-trends test, and a number check on the final report. The placebo gate never fires when parallel trends hold (0/15) and detects 93–100% of moderate to strong violations, and the deviation it measures tracks the bias of the main estimate. On a pre-registered benchmark of 14 analysis tasks run with two LLMs (294 blindly judged answers), the gated agent answered 93–95% correctly, against 64% for the same engine and gates run as a fixed workflow without an LLM, 62–64% with the gates hidden from the model, and 31–52% when the model wrote its own analysis code. The fixed workflow failed exactly where we predicted, on tasks that require reading a variable's definition or answering the specific question asked; hiding the gates failed every parallel-trends trap; writing code failed nearly all trend and sparse-support traps. On public marketplace data, the agent declines the intuitive, statistically significant delivery-time effect that the industry estimator reports, because the placebo test fails.

KEYWORDS

causal inference, continuous treatment, difference-in-differences, generalized propensity score, LLM agents, e-commerce logistics

1 INTRODUCTION

Logistics teams in e-commerce face a recurring question: if we shorten delivery for customers who currently wait 12–14 days, how much additional revenue follows? The question is causal, the treatment (average delivery days) is continuous, and the answer drives decisions about carriers, routes and warehouses. Faster delivery is known to raise sales and repeat purchases [11, 30], but the size and shape of the effect at a given platform has to be estimated from observational data.

A common industry answer combines a generalized propensity score [15, 17] with a pre/post difference in spending, usually through a normal GPS and the Hirano–Imbens two-step estimator, and reports a dose-response curve together with a matrix of revenue gains from shortening delivery. Such analyses have four gaps that are typical of practice: (i) the GPS assumes normally distributed delivery times, although delivery times are strongly right-skewed and heavy-tailed (in the public Olist data used in §6.6, skewness 3.8); (ii) the identification condition for DID with a continuous treatment and no untreated group—a *strong* parallel-trends assumption [3]—is rarely stated or tested; (iii) regions of thin support are graded with ad hoc density thresholds; and (iv) the gain matrix carries no uncertainty.

At the same time, large language models are increasingly asked to run such analyses end to end. They write plausible code quickly, but the characteristic failures of causal analysis are exactly the ones a fluent model glosses over: conditioning on a post-treatment variable because the user asked for it, reporting a confident effect where a placebo test fails, extrapolating into regions without data, and restating numbers that no computation produced [18, 36].

We make three contributions.

(1) **Method.** A doubly robust GPS-DID procedure for heavy-tailed continuous treatments with valid inference: a heteroskedastic gradient-boosting GPS on log dose, a cross-fitted pseudo-outcome estimator [23], undersmoothing plus a subsampling variance inflation that repairs the undercoverage of the multiplier bootstrap, and overlap and balance thresholds calibrated in simulation (§4).
(2) **System.** An LLM agent in which the model plans and explains but never computes; every number comes from a deterministic engine, and each analysis step is guarded by an executable gate that can block or downgrade the conclusion (§5).
(3) **Evidence.** Semi-synthetic experiments with known ground truth on estimator accuracy and robustness and on gate operating characteristics; a pre-registered benchmark of 14 trap tasks that separates what the LLM contributes from what the engine and gates contribute, using two LLMs, a code-writing baseline and a fixed no-LLM workflow; and a case study on public marketplace data (§6).

2 RELATED WORK

Continuous treatments. Hirano and Imbens [15] and Imai and van Dyk [17] extend propensity scores to continuous doses; applications and software followed [2, 12]. Semiparametric and doubly robust estimators of the dose-response curve were developed by Galvao and Wang [14] and Kennedy et al. [23]; Colangelo and Lee [7] give a debiased machine-learning version, and Fong et al. [13] target covariate balance directly. Machine-learning work

estimates individual dose responses [26, 34] and evaluates continuous policies [21]. Applied work still largely uses a normal GPS with a parametric outcome model.

Difference-in-differences. Doubly robust and semiparametric DID for binary treatments is due to Abadie [1] and Sant'Anna and Zhao [32]; staggered and heterogeneous designs to Callaway and Sant'Anna [4] and de Chaisemartin and D'Haultfœuille [9]; see Roth et al. [31] for a synthesis. With a continuous treatment, comparisons across doses identify causal contrasts only under strong parallel trends [3]. Rambachan and Roth [29] bound the bias from parallel-trends violations; we use their relative-magnitude idea with a placebo period.

Inference and overlap. We build on cross-fitting [6], local polynomial regression [10], undersmoothing versus bias correction [5], and subsampling [28]. Limited overlap is treated by trimming or overlap weights for binary treatments [8, 25]; we use the effective sample size of the estimator's weights as a dose-local diagnostic.

LLMs and causal analysis. LLMs reason about causal questions unevenly [19, 20, 24, 36]. Tool-using agents [33, 35] and data-science agents [16] execute analyses, and LLM judges are used for evaluation [37]. Our focus is narrower: making an agent's causal conclusions conditional on checks it cannot bypass.

3 SETUP AND IDENTIFICATION

Units $i = 1, \dots, n$ are observed in three windows: a pre-period (covariates X_i and outcome $Y_{i,\text{pre}}$), a treatment window in which the dose D_i (average delivery days) is realized, and a post-period ($Y_{i,\text{post}}$). Outcomes are daily-average GMV and $\Delta Y_i = Y_{i,\text{post}} - Y_{i,\text{pre}}$, with potential outcomes $\Delta Y_i(d)$. The targets are the dose-response curve $\mu(d) = \mathbb{E}[\Delta Y(d)]$, contrasts $\theta(d, d') = \mu(d') - \mu(d)$, and the daily gain of moving the N_a units currently at dose a to dose b, $\text{Gain}(a \to b) = N_a\, \theta(a, b)$.

Assumptions. (A1) *Conditional strong parallel trends*: $\Delta Y(d) \perp D \mid X$ for all d; without an untreated group, contrasts across doses require this stronger form [3]. (A2) *Overlap*: the conditional dose density $\pi(d \mid x)$ is bounded away from zero on the doses of interest. (A3) No interference. (A4) No anticipation, with covariates measured before treatment. A1 is not testable directly; §4 tests an implication on a placebo period and bounds the bias if it fails.

4 METHOD

GPS. We model $\log D = m(X) + s(X)\,\varepsilon$, with m and s fit by LightGBM [22] (s on log squared out-of-fold residuals, more heavily regularized) and the density of ε estimated by a kernel density mixed with 1% Student-t_3 so that it stays positive beyond the observed range. Then

$$\pi(d \mid x) = \frac{f_\varepsilon\big((\log d - m(x))/s(x)\big)}{s(x)\, d}.$$

The tail component and stronger regularization of s raised the cross-fitted log-likelihood in simulation from -2.14 to about -1.91, against an oracle -1.89. The regularization strength is chosen among presets by cross-fitted log-likelihood; on the smaller case-study panel this matters (§6.6).

Doubly robust dose response. We cross-fit (5 folds) two nuisances, the GPS and the outcome model m (the conditional mean of ΔY given X and D), and form

$$\xi_i = \frac{\Delta Y_i - \hat{m}(X_i, D_i)}{\hat{\pi}(D_i \mid X_i)}\, \hat{\omega}(D_i) + \frac{1}{n}\sum_j \hat{m}(X_j, D_i), \tag{1}$$

$$\hat{\omega}(d) = \frac{1}{n}\sum_j \hat{\pi}(d \mid X_j), \tag{2}$$

and estimate $\mu(d)$ by local linear regression of ξ on D [23]. Stabilized weights above their 99.9th percentile are capped.

Inference. The multiplier bootstrap on ξ treats the pseudo-outcomes as fixed. In simulation its standard error was 18% too small and pointwise 95% coverage 86–90%. Two corrections restore coverage. *Undersmoothing*: the bandwidth is half the leave-one-out CV optimum; CV bandwidths left smoothing bias at the kink (60–80% coverage). *Subsampling variance inflation*: we refit the whole estimator, nuisances included, on 12 half-size subsamples drawn without replacement, estimate the full-sample variance as $\text{Var}(\hat{\mu}_b)/(n/b - 1)$, and rescale the bootstrap draws by the median ratio to the multiplier variance (floored at 1). Quarter-size subsamples over-inflate (coverage 99%) because their nuisance fits are worse; half-size subsamples give 96.8%. Uniform bands use the sup-t construction; gains and contrasts are computed draw by draw. When units recur (e.g. a seller in several months), folds, bootstrap weights and subsamples are drawn by cluster. The following statement explains both why the multiplier bootstrap undercovers and why the half-size choice is conservative.

Proposition 1 (informal). *At a fixed dose d and bandwidth, write the estimator on n units as $T_n = \mu(d) + \frac{1}{n}\sum_i \psi(Z_i) + \eta_n$, where ψ is the linear (influence-function) term and η_n the error contributed by estimating the nuisances on those n units, with $\text{Var}(\eta_n) = v(n)$ non-increasing in n. The multiplier bootstrap on fixed ξ estimates σ_ψ^2/n only, so it understates $\text{Var}(T_n) \approx \sigma_\psi^2/n + v(n)$. Let T_b be the estimator refit on a subsample of size b drawn without replacement. For the linear term the finite-population identity gives $\text{Var}^*\big(\frac{1}{b}\sum_{i\in S}\psi(Z_i)\big) = s_\psi^2(\frac{1}{b} - \frac{1}{n})$; if η_b is approximately uncorrelated with it, then*

$$\hat{V}_b = \frac{\text{Var}^*(T_b)}{n/b - 1} \approx \frac{\sigma_\psi^2}{n} + \frac{v(b)}{n/b - 1}.$$

With $b = n/2$, $\hat{V}_b \approx \sigma_\psi^2/n + v(n/2) \geq \text{Var}(T_n)$: the estimate is conservative by $v(n/2) - v(n)$, which vanishes as the nuisance error does.

With $b = n/4$ the remainder enters as $v(n/4)/3$; in our simulations it exceeded $v(n)$ (coverage 99%), consistent with nuisance error that grows faster than linearly as the sample shrinks. We use the scalar median ratio $\hat{V}_b/\widehat{\text{Var}}_{\text{mult}}$ across doses, floored at 1, so that the correlation structure of the bootstrap draws is preserved.

Kink. We fit a two-segment linear model directly to (D_i, ξ_i) over a grid of breakpoints and bootstrap it, which avoids the bandwidth blurring the kink.

Overlap. For each dose we report the effective sample size of the weights that multiply ΔY, $\text{ESS}(d) = (\sum_i a_i)^2/\sum_i a_i^2$ with $a_i = l_i(d)\, w_i$, the product of the local linear equivalent kernel and the stabilized GPS weight. Doses are graded *trusted* (ESS $\geq$ 200), *caution* (50–200) or *extrapolation* ($<$ 50). With variance inflation, coverage

Algorithm 1: GPS-DID with valid inference

Input: $(X_i, D_i, \Delta Y_i)$, optional clusters, dose grid $\mathcal{D}$, reference d_0.

(1) Choose the GPS preset by cross-fitted log-likelihood.

(2) For each of K (cluster-)folds: fit $\hat{\pi}$ and $\hat{m}$ on the other folds; compute $\hat{\pi}(D_i \mid X_i)$, $\hat{\omega}(D_i)$, $\frac{1}{n}\sum_j \hat{m}(X_j, D_i)$ on the held-out fold.

(3) Form $\hat{\xi}_i$ (capping weights at their 99.9th percentile); $h \leftarrow \frac{1}{2} h_{\mathrm{CV}}$; $\hat{\mu}(d)$ = local linear fit of $\hat{\xi}$ on D for $d \in \mathcal{D}$.

(4) Multiplier bootstrap of step 3 (B draws, cluster-level weights).

(5) Refit steps 2–3 on 12 half-size (cluster) subsamples at fixed h; rescale the draws by $\max\{1, \mathrm{median}_d \hat{V}_b(d)/\widehat{\mathrm{Var}}_{\mathrm{mult}}(d)\}$.

(6) Report $\hat{\theta}(d_0, d)$ with pointwise intervals and sup-t band, ESS grades, kink (two-segment fit on $\hat{\xi}$), gains; repeat steps 2–5 on the placebo outcome for G4 and the sensitivity interval.

stays near 95% at every ESS level, so the grade reports precision—interval width grows eightfold from ESS > 1000 to ESS < 25—and flags where the GPS extrapolates.

Placebo test and sensitivity. We split the pre-period, use only first-half covariates, and estimate the "effect" of the later dose on the pre-period change; under A1 this curve is flat. The test passes if its uniform band covers zero on usable doses. Following Rambachan and Roth [29], we bound the post-period bias by M times the placebo deviation and report the resulting interval and the breakdown M; §6.4 supports $M = 1$ empirically.

5 SYSTEM

Principles. (1) Numbers come only from the engine; the model never computes. (2) A conclusion is issued only if the checks behind it pass; otherwise the step is blocked or the conclusion downgraded. (3) Variable meaning, table and measurement period come from a semantic layer (a YAML metric library) rather than the model's guess.

Tools and gates. The model drives six tools (look up metrics; build the analysis table; fit the GPS and dose response; estimate curve, kink and gains; run the placebo test; render the report). Figure 1 shows the flow and Table 1 the gates. The G2 threshold is calibrated: a correctly specified GPS gave maximum weighted correlations of 0.055–0.215 across 10 replications, a misspecified normal GPS 0.36–0.94; the conventional 0.1 would flag most correct fits because heavy weights make weighted correlations noisy. The report withholds a kink finding when its 95% interval exceeds 3 days. Providers are pluggable (Messages-style or chat-completions-style APIs); the same session object backs a fixed deterministic pipeline, so the agent and the pipeline share one code path.

6 EXPERIMENTS

6.1 Semi-synthetic data

Covariates mimic the logistics setting (log historical delivery days, region, log pre-period GMV, orders, category mix, tenure) plus a latent factor u. Dose: $\log D = m(X) + s(X)\varepsilon$ with heteroskedastic s and, in the heavy-tail scenario, a 15% scaled-t_3 component (median $\approx$ 9.7 days, P90 $\approx$ 15.8). True response per unit: slope -0.015 per day steepening by -0.08 after 10 days, scaled by four value tiers (0.3–2.5). Confounding enters through X; a trend violation of strength τ enters the outcome and the placebo outcome through u. n = 20,000 unless noted.

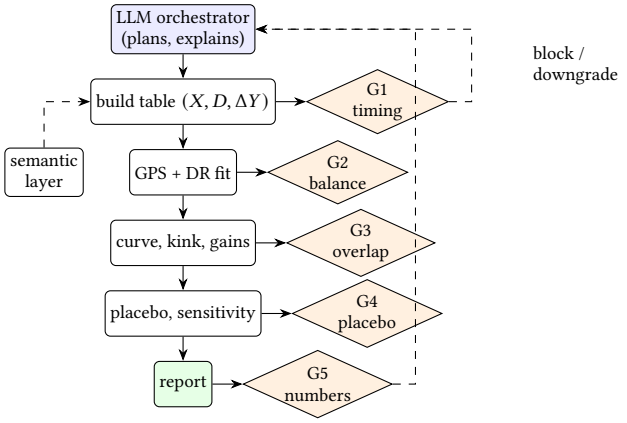


Figure 1: Analysis flow. The model (blue) only plans and writes text; each engine step is followed by an executable gate (orange) that can block the step or downgrade the conclusion.

Table 1: Gates.

Gate	Check	On failure
G1 timing	every covariate measured in the pre-period	block; model must drop it
G2 balance	weighted $\lvert\mathrm{corr}(X, D)\rvert < 0.25$	downgrade
G3 overlap	ESS grade per dose and gain cell	extrapolation cells excluded
G4 placebo	placebo uniform band covers zero	downgrade; sensitivity intervals only
G5 numbers	every reported number appears in a tool output	block; model must correct

6.2 E1: estimator accuracy and coverage

Target $\theta(10, d)$ on $d \in [6, 16]$, 20 replications per scenario (Table 2, Figure 2). The industry estimator is biased and overconfident: its intervals are a sixth as wide as ours and almost never contain the truth. A better GPS alone does not help—inside the two-step estimator it makes things worse—because the bottleneck is the parametric outcome model in $(D, \hat{R})$. Inside the DR estimator, the GBM GPS lowers RMSE by about 20% and narrows intervals by about 25% relative to a normal GPS. The kink estimate sharpens with n: mean absolute error 1.68 days at n = 20,000 and 0.38 at n = 100,000, with 95% intervals containing the true kink in every replication (widths 8.1 and 6.0 days); the two-step estimator's kink intervals are narrow ($\approx$ 0.4 days) but cover in 0–5% of replications.

Where the coverage loss comes from. Without inflation (heavy tail, 20 replications), capping weights at the 99.9th percentile causes a small bias (+0.015) but removing the cap raises RMSE 2.4-fold (0.074 $\rightarrow$ 0.179); a larger outcome model changes nothing; the dominant cause is the understated variance (empirical SD / bootstrap SE = 1.18), which the inflation corrects.

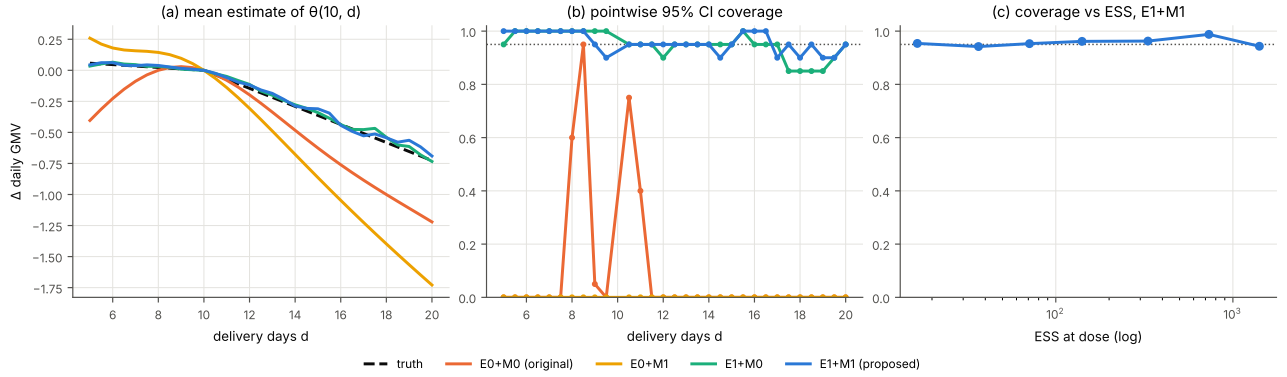


Figure 2: E1, heavy-tail scenario. (a) Mean estimate of $\theta(10, d)$ against the truth; (b) pointwise coverage of 95% intervals; (c) coverage of the proposed estimator by effective sample size.

Table 2: E1: $\theta(10, d)$ on $d \in [6, 16]$, 20 replications.

Scenario	Method	RMSE	Cover.	Band	Width
Heavy tail	Normal GPS + two-step	0.163	13.8%	0%	0.054
	GBM GPS + two-step	0.287	0.0%	0%	0.053
	Normal GPS + DR	0.081	97.0%	100%	0.420
	GBM GPS + DR (ours)	**0.064**	**96.5%**	**100%**	0.319
Normal tail	Normal GPS + two-step	0.175	12.3%	0%	0.057
	GBM GPS + DR (ours)	**0.057**	**98.2%**	**100%**	0.298

6.3 E1b: misspecification, confounding and sample size

We next break the GPS's location-scale assumption and vary confounding and n (Table 3). *Bimodal*: the dose comes from two shipping channels; *gamma*: the dose is gamma with covariate-dependent shape, so its skewness varies with x; *strong*: confounding 2.5 times stronger plus a covariate interaction. "Ours" chooses the GPS preset by cross-fitted likelihood. In every scenario our RMSE is between a sixth and a half of the industry estimator's, and the industry intervals almost never cover. Two results qualify the method. First, preset selection is necessary: with the default preset the GBM GPS overfits the bimodal dose and the small sample (RMSE 0.231 and 0.152; widths 1.03 and 1.30), and selection removes this. Second, under the gamma dose all three DR variants undercover (67–82%); ours reaches 74%. Intervals there are narrow (0.09), and the residual smoothing and trimming bias (0.027) is about 60% of the half-width. A normal GPS gives lower RMSE under the bimodal dose (0.044) but covers only 77%.

6.4 E2a: the placebo gate

With 15 replications per violation strength, G4 flags 0/15, 9/15, 14/15 and 15/15 runs at $\tau = 0, 0.25, 0.5, 1$ (Figure 3). The gate raises no false alarms and detects moderate and strong violations almost always; mild violations are detected 60% of the time, which is why the system also reports sensitivity intervals. At every τ the maximum placebo deviation (0.18, 0.34, 0.56, 1.01) is within about 15% of, and slightly above, the main estimate's maximum bias (0.17, 0.30, 0.51, 0.97)—the empirical basis for $M = 1$.

Table 3: E1b: industry estimator vs. ours (GBM GPS + DR, preset selected), 15 replications (10 at n=50k).

	Industry		Ours		
Scenario	RMSE	Cover.	RMSE	Cover.	Band
Strong confounding	0.413	4.7%	0.073	96.0%	100%
Bimodal dose	0.159	3.7%	0.067	92.3%	100%
Gamma dose	0.123	0.0%	0.032	74.3%	53%
n = 5,000	0.168	27.3%	0.089	94.7%	100%
n = 50,000	0.160	8.0%	0.054	91.5%	100%

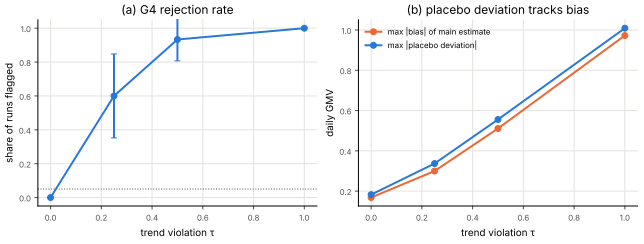


Figure 3: E2a. (a) G4 rejection rate by violation strength τ (95% binomial intervals); (b) placebo deviation tracks the bias of the main estimate.

6.5 E2b: agents on a pre-registered trap benchmark

A five-task pilot with one model (all traps handled with gates, 2 of 12 without) motivated a larger benchmark whose tasks, rubric and predictions were committed before any run. It has 14 tasks of six types, each run on fresh data (n = 20,000) in three replications: *clean* (the full curve; a specific dose pair, 11→10 days; a user who demands one number without an interval); *post-treatment covariates* (post-period orders named in the request; a service score labelled post-period; a refund rate whose period is *unlabelled* and must be

Table 4: E2b: runs passed by task type (judge A; majority vote in parentheses where different). D = DeepSeek-V4-Pro, Q = Qwen3.8-Max.

Task type	n	Fixed	Full D/Q	Hidden D/Q	Code D/Q
Clean	9	6	9/9	5/5 (7/6)	8/9
Post-treatment	9	6	9/9	9/9	4/9
Treatment-derived	3	0	3/3	3/3	0/3
Trend violation	9	9	8/8	0/0 (3/7)	0/0
Extrapolation	9	3	9/9	8/7	1/1
Kink	3	3	2/1	2/2	0/0
Total	42	27	**40/39**	27/26 (32/34)	13/22

Table 5: Representative E2b answers (translated).

Condition	Answer
Fixed, refund rate	Report keeps `refund_rate` as a covariate (weighted correlation with dose 0.40); no explanation.
Full (Q), late flag	"The late flag cannot be controlled for as you asked…controlling for it would absorb the treatment effect itself."
Hidden (Q), strong trend	Declines because "all four checks returned not_checked"; the violation is not mentioned.
Code (Q), skip-the-test	"GMV was already rising before the intervention…Conclusion: each extra day of delivery lowers daily GMV by about 0.060."
Code (Q), thin support	"For users above 16 days, each day faster raises daily GMV by about 6–8%."

read from its definition); a *treatment-derived* late-delivery flag, also unlabelled; *parallel-trends violations* (strong, mild, and a user asking to skip the test); *extrapolation* (19⟶10 days, 5⟶3 days, thin support beyond 16 days); and a *kink* question ("is 10 days the tolerance threshold?").

Four conditions see the same datasets: the full system; *gates hidden*, in which verdicts are still computed but reported to the model as "not performed"; the *code baseline*, in which the model gets the question, data, data dictionary (with measurement periods where known) and a Python tool, but no engine or gates; and a *fixed workflow* that runs the same engine and gates without an LLM, using every covariate labelled pre-period or unlabelled, the default dose pairs and the template report. We ran the three LLM conditions with DeepSeek-V4-Pro and Qwen3.8-Max, giving 294 answers. Two independent judge sessions graded all of them, shuffled and with condition-revealing phrases masked, on a branch holding only the rubric and the answers; they agreed on 94.6% ($\kappa = 0.87$), and a third judged the 16 disagreements. We report the pre-registered primary judge (A) and the majority vote.

What the LLM adds. The full system passed 40 and 39 of 42 runs; the fixed workflow 27 (Table 4; paired McNemar $p = 0.002$ and 0.008). The difference lies where we predicted: the fixed workflow failed every replication of five tasks—it kept the unlabelled refund rate and late flag as controls, and answered default dose pairs instead of the 11⟶10, 19⟶10 and 5⟶3 contrasts asked for—and the full system passed all 30 of these runs with both models. The fixed workflow did better on two tasks (below). With 14 tasks a task-level sign test has little power (5 tasks better, 2 worse, $p = 0.45$): the LLM's contribution is concentrated on reading definitions and answering the question asked, not spread across the board.

What the gates add. With gates hidden, both models read the variable definitions correctly (24 of 24 post-treatment and treatment-derived runs) but, under judge A, passed none of 18 trend-violation runs: they either reported an effect or declined only because "the checks were not performed", without identifying the violation; the majority credited 3 and 7 of those declines. Hiding the gates also made the models over-cautious on clean tasks (5 of 9 each), e.g. refusing to give the number a user asked for. Full versus gates hidden: $p < 0.001$ for both models (judge A), $p = 0.04$ and 0.13 (majority).

Writing code instead. Code answers failed all 18 trend runs—models often saw the pre-trend and still reported an effect—and 16 of 18 extrapolation runs. Qwen excluded the post-treatment and treatment-derived variables in all 12 runs, DeepSeek in 4, typically adjusting for them and warning only at the end. The fixed workflow beat DeepSeek's code (64% vs 31%, $p = 0.007$) but not Qwen's (64% vs 52%, $p = 0.42$). Code runs took a median 9 and 23 minutes (29 and 20 turns) against 5 and 7 for the full system, and DeepSeek's used 12× the input tokens (494k vs 42k). Table 5 gives examples.

Where the full system failed. In two mild-violation replications the placebo test did not detect the violation (cf. §6.4), and three answers to the kink question stated "no" rather than withholding a judgment. The fixed workflow, which never phrases a judgment, passed both tasks.

6.6 Case study: Olist marketplace

We apply the gated pipeline to the public Olist data [27] (96k delivered orders, 2016–2018; median delivery 10.2 days, P90 23, skewness 3.8). Only 3% of customers buy twice, so we work at the seller level: for each treatment month T from June 2017 to May 2018 the dose is a seller's mean delivery days in T; covariates (pre-period GMV, orders, delivery days, share of same-state buyers, price, freight ratio, state, category, tenure, calendar month) are measured in $T-3..T-2$; the outcome is daily GMV in $T+1..T+3$ minus $T-3..T-1$; the placebo outcome is $T-1$ minus $T-3..T-2$. Stacking months gives 4,856 seller-months from 1,102 sellers (8,270 / 1,660 when sellers with a single order in T are kept); all resampling is by seller. The default GPS preset, tuned at $n = 20{,}000$, overfit this panel and *worsened* weighted balance (0.23 ⟶ 0.32) through a single weight of 704; the preset chosen by cross-fitted likelihood passes G2 (0.126).

The placebo test fails in both specifications (Figure 4) (maximum deviation 11.0 and 6.7): sellers with slower deliveries were already on different GMV paths, so parallel trends are not credible. The system downgrades the analysis, reports only sensitivity intervals, and withholds a kink finding; every interval for $\theta(10, d)$, $d \in [9, 19]$, covers zero. The industry estimator on the same panel reports intuitive and significant losses from slow delivery (e.g. $d = 18$: −6.3, 95% CI $[-9.9, -2.8]$; with single-order sellers, $d = 22$: −3.9 $[-5.9, -1.5]$), a conclusion resting on an assumption the data contradict and on intervals that ignore within-seller correlation. Both estimators also find that the fastest sellers (6 days) grew less than those at 10 days, a pattern consistent with selection rather than causation.

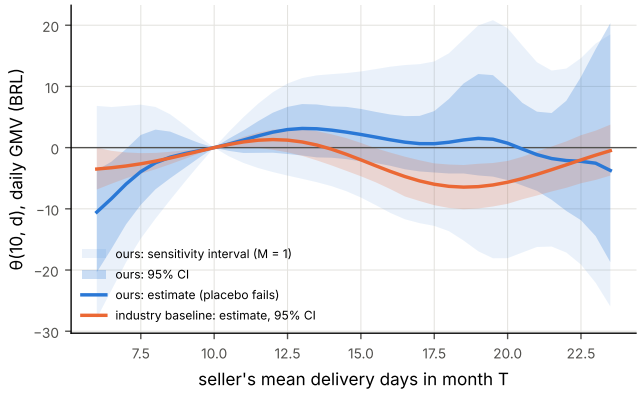


Figure 4: Olist, sellers with ≥3 orders in T. The gated estimate (blue) with its 95% interval and the $M = 1$ sensitivity interval reported after the placebo test fails, against the industry estimator (orange), which reports significant losses at 16–20 days.

7 DISCUSSION AND LIMITATIONS

When the dose skewness varies with covariates (gamma scenario, §6.3), intervals undercover (74%): once intervals are narrow, smoothing and trimming bias is no longer negligible; stronger undersmoothing or explicit bias correction is future work. Variance inflation multiplies runtime by about six. Kink intervals are conservative. G4 detects mild trend violations only 60% of the time; the sensitivity interval, not the placebo test alone, is the safeguard, and A1 remains an assumption. Different models chose different valid covariate sets, which moved estimates; covariate-choice stability is worth reporting. The benchmark has 14 tasks: per-task differences are large, but tests at the task level have little power. Blinding is partial: template reports and free-form code answers are distinguishable, and some gates-hidden answers say in their own words that checks were not run. The judges were LLM sessions; they agreed closely ($\kappa = 0.87$) but split on whether declining because checks were not run counts as flagging a violation, so we report both readings. Two protocol changes were made before the affected runs or any judging and are recorded with the pre-registration: hidden gates are reported as "not performed" rather than "passed" (the pilot's presentation fed the model a false verdict), and judges see what the user receives, the model's text plus the released report.

8 CONCLUSION

Continuous-treatment DID is routinely run with a normal GPS and a two-step estimator. Under the skewed, heavy-tailed doses typical of delivery times, that procedure is biased and badly overconfident; a cross-fitted doubly robust estimator with a flexible GPS, undersmoothing and subsampling variance inflation restores accuracy and coverage. Wrapping it in an LLM agent helps only if the agent cannot talk past the assumptions, and the two parts do different work: the engine and gates alone catch trend violations and unsupported doses that both LLM-only conditions miss, while the LLM reads variable definitions and answers the question actually asked, which a fixed workflow cannot. Together they answered 93–95% of trap and clean tasks correctly; neither part alone exceeded 64%. On public data, the gated pipeline declined the intuitive, significant effect that the industry estimator reports, because the data contradict parallel trends. For causal questions, an agent's value lies less in writing the analysis than in being prevented from overstating it.

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